

Setting. Let $A \in \mathbb{R}^{n \times d}$ be partitioned into row windows $A_1, \dots, A_W$ processed one at a time. At window w the state carried from earlier windows is a sketch $B \in \mathbb{R}^{r_{\text{sk}} \times d}$ and a row sample (reservoir) $R \in \mathbb{R}^{r_{\text{R}} \times d}$. In the lookahead variant developed here, the state also includes a fixed-size future buffer F_w containing a contiguous row block observed after the current block with a lag of $\text{row}(F_w)$ rows. Let

$$N_w := \begin{pmatrix} A_1 \\ \vdots \\ A_w \end{pmatrix}, \quad C_w := \begin{pmatrix} R \\ A_w \end{pmatrix}, \quad L_w := \begin{pmatrix} A_w \\ F_w \end{pmatrix}, \quad M_w := \begin{pmatrix} B \\ A_w \end{pmatrix}. \quad (1)$$

For a nonzero y , the Rényi-2 / inverse-participation entropy is

$$h_2(y) := -\log\left(\frac{\|y\|_4^4}{\|y\|_2^4}\right) = 4 \log \|y\|_2 - 4 \log \|y\|_4, \quad (2)$$

bounded between 0 (one-spike) and $\log \dim(y)$ (uniform). Throughout $v \in \mathbb{S}^{d-1}$; assume $\text{row}(C_w) > 1$, $\text{row}(L_w) > 1$, and $\text{row}(N_w) \leq n$.

The current window A_w and the lookahead buffer F_w are disjoint row sets. Setting $F_w = \emptyset$ and ignoring assumption (A1') below recovers a pure past-state online method driven by the additive score alone.

Assumptions. Two past-state approximations:

$$\|Bv\|_2^2 + \|A_w v\|_2^2 \approx \|N_w v\|_2^2, \quad (\text{A1})$$

$$\frac{h_2(C_w v)}{\log \text{row}(C_w)} \approx \frac{h_2(N_w v)}{\log \text{row}(N_w)} \approx \frac{h_2(Av)}{\log n}. \quad (\text{A2})$$

(A1) says the sketch preserves the aggregate past-row energy; (A2) says the pooled reservoir-plus-current entropy tracks the full-matrix entropy at the same rate.

For the future-consistency variant, add a stationarity assumption on per-row squared action:

$$\frac{\|A_w v\|_2^2}{\text{row}(A_w)} \approx \frac{\|F_w v\|_2^2}{\text{row}(F_w)} \approx \frac{\|Av\|_2^2}{n} =: \lambda(v). \quad (\text{A1}')$$

(A1') says that v 's per-row gain is the same on the current window, on the immediately-following lookahead rows, and on the full matrix. It is strictly stronger than (A1): (A1) requires only additive energy conservation of the sketch, while (A1') requires rate stationarity across row blocks. (A1') is plausible when v captures genuinely repeated structure across row blocks and is violated by directions whose support lives in a single window.

Resulting inequality. Under (A2) and $\|N_w v\|_4 \leq \|Av\|_4$,

$$\|Av\|_2^2 \gtrsim \|N_w v\|_2^2 \left(\frac{\text{row}(N_w)}{n} \right)^{-h_2(N_w v)/(2 \log \text{row}(N_w))}. \quad (\star)$$

The exponent $-h_2/(2 \log \text{row})$ is an entropy-dependent subsampling correction: if the response $N_w v$ is diffuse (large h_2), the subsample gain closely approximates the full-matrix gain; if it is concentrated (small h_2), the correction factor rises, reflecting the possibility that subsampling may miss concentrated support rows.

Additive combined score. Substituting (A1) and (A2) into $(\star)$ gives

$$\|Av\|_2^2 \gtrsim (\|Bv\|_2^2 + \|A_w v\|_2^2) \phi_w(v) =: \text{score}_w^{\text{add}}(v), \quad (3)$$

with

$$\phi_w(v) := \left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2} \right)^{c_w}, \quad c_w := \frac{\log(\text{row}(N_w)/n)}{2 \log \text{row}(C_w)}. \quad (4)$$

The gain factor $\|Bv\|_2^2 + \|A_w v\|_2^2$ is a proxy for $\|N_w v\|_2^2$ by (A1). Since $\text{row}(N_w) \leq n$, $c_w \leq 0$, and high-entropy pooled vectors are boosted by the correction. If no old row sample is available, take $R = \emptyset$; if no old sketch is available, take $B = 0$.

Future-consistency harmonic-mean score. Define per-row rate estimators

$$\hat{\lambda}_{A_w}(v) := \frac{\|A_w v\|_2^2}{\text{row}(A_w)}, \quad \hat{\lambda}_{F_w}(v) := \frac{\|F_w v\|_2^2}{\text{row}(F_w)}, \quad (5)$$

and the harmonic mean

$$\text{HM}(x, y) := \frac{2xy}{x + y + \varepsilon}. \quad (6)$$

Under (A1'), $\hat{\lambda}_{A_w}(v) \approx \hat{\lambda}_{F_w}(v) \approx \lambda(v)$, so any symmetric mean of the two recovers $\lambda(v)$. We choose the harmonic mean because it penalises violations of (A1') more aggressively than the arithmetic or geometric mean: $\text{HM}(x, y) \leq \text{GM}(x, y) \leq \text{AM}(x, y)$, with equality only when $x = y$, and $\text{HM}(x, y) \rightarrow 0$ whenever either argument vanishes while AM remains nonzero. This matches the failure mode the HM variant is designed to reject, in which a direction is highly aligned with the current block but has no future support.

The resulting score is

$$\text{score}_w^{\text{HM}}(v) := n \cdot \text{HM}(\hat{\lambda}_{A_w}(v), \hat{\lambda}_{F_w}(v)) \cdot \phi_w(v), \quad (\text{HM-SCORE})$$

equivalently

$$\text{score}_w^{\text{HM}}(v) = \frac{2n \|A_w v\|_2^2 \|F_w v\|_2^2}{\text{row}(F_w) \|A_w v\|_2^2 + \text{row}(A_w) \|F_w v\|_2^2 + \varepsilon} \cdot \phi_w(v). \quad (7)$$

Under (A1'), $n \cdot \text{HM}(\hat{\lambda}_{A_w}, \hat{\lambda}_{F_w}) \approx n\lambda(v) = \|Av\|_2^2$, so

$$\|Av\|_2^2 \approx \text{score}_w^{\text{HM}}(v) \cdot \phi_w(v)^{-1} \quad \text{under (A1')}, \quad (8)$$

and the score itself equals $\|Av\|_2^2 \cdot \phi_w(v)$ under (A1'). The HM-score is therefore an *estimator* of $\|Av\|_2^2 \cdot \phi_w(v)$ that is conservative against (A1') violations, not a proof-style lower bound.

[INCOMPLETE] Deriving the HM form from the subsampling inequality ($\star$). The additive score inherits the lower-bound interpretation $\text{score}_w^{\text{add}}(v) \leq \|Av\|_2^2$ directly from ($\star$) + (A1) + (A2). A corresponding derivation that yields $\text{score}_w^{\text{HM}}(v)$ as a bound on $\|Av\|_2^2$ starting from ($\star$) (or a Hölder-class relative) is *not written down*. Required pieces that are missing:

- A subsampling inequality analogous to ($\star$) but applied symmetrically to two disjoint row blocks A_w, F_w of A rather than to a nested prefix $N_w \subset A$.
- A probabilistic or deterministic argument under which the harmonic mean of per-row rates lower-bounds $\|Av\|_2^2/n$. Candidate tools: a Chebyshev / second-moment argument treating $\hat{\lambda}_{A_w}$ and $\hat{\lambda}_{F_w}$ as independent estimators of $\lambda(v)$, a matrix concentration bound on row partitions, or a direct rearrangement inequality converting $\text{HM} \leq \text{GM} \leq \text{AM}$ into a two-sided bound on $\lambda(v)$ with a controlled gap.
- Handling of the lookahead lag. The online variant accesses F_w only after a fixed lag, so any bound must account for the residual rows between A_w and F_w that are neither used nor available, or relate $\lambda(v)$ computed on the two windows to $\lambda(v)$ on the full matrix under mixing/stationarity assumptions.

Until this derivation is filled in, $\text{score}_w^{\text{HM}}$ should be understood as a stationarity-based estimator with empirically verified robustness, not as a theorem-backed lower bound. The additive form remains the reference when a proof-style bound is required.

Algorithm overview. Process one row-window at a time. For the first window, there is no previous sketch and no old row sample, so $B = 0$, $R = \emptyset$, $C_1 = A_1$, $M_1 = A_1$, and (in the lookahead variant) F_1 is taken from the rows immediately following A_1 . For a later window w , the input state is the carried sketch $B^{(w-1)} = \Sigma_{w-1} V_{w-1}^\top$, the row sample $R^{(w-1)}$, and the lookahead buffer F_w . The algorithm at window w performs three stages:

1. *Stage 1 — subspace selection.* Evaluate score_w (either form) on a finite candidate pool $\mathcal{P}_w$ and pick the r -element subset with largest total score. Let $\mathcal{S}_w$ denote the span of the selected candidates.
2. *Stage 2 — Rayleigh–Ritz frame refinement.* Let Q_c be an orthonormal basis of $\mathcal{S}_w$ (via QR). Compute the restricted SVD $M_w Q_c = U_c \Sigma_c V_c^\top$ and take $\hat{V}_w := Q_c V_c$.

3. *Stage 3 — carry-sketch update.* Form $Y_w := M_w \widehat{V}_w$, $U_{\cdot,w} := \text{orth}(Y_w)$, and compute $U_{\cdot,w}^\top M_w = U_{\text{small},w} \Sigma_w V_w^\top$. The sketch passed to the next window is $B^{(w)} = \Sigma_w V_w^\top$. The row sample $R^{(w)}$ is updated from the raw rows seen through window w .

The rest of the document expands and justifies each stage.

Stage 1 — subspace selection by simple argmax. Let score_w denote either the additive or the HM score. Stage 1 is a single argmax over size- r subsets of the pool:

$$\tilde{V}_w \in \arg \max_{S \subseteq \mathcal{P}_w, |S|=r} \sum_{v \in S} \text{score}_w(v), \quad \mathcal{S}_w := \text{span}(\tilde{V}_w). \quad (\text{S1})$$

When pool elements are linearly independent and distinct in score, (S1) reduces to top- r -by-score selection. No orthogonality constraint is imposed in Stage 1 and no sequential greedy structure is needed. Orthogonality is handled by the Rayleigh–Ritz reframe in Stage 2, which operates on any basis of $\mathcal{S}_w$. This simplification is valid whenever the pool elements are structurally distinct enough that the top- r span $\mathcal{S}_w$ is genuinely r -dimensional; the Stage-2 SVD flags the degenerate case through $\sigma_r(M_w Q_c) \rightarrow 0$, and the algorithm can fall back to a smaller effective rank.

Candidate pool. The pool $\mathcal{P}_w$ is a fixed set of $|\mathcal{P}_w| \geq r$ online-computable unit vectors in $\mathbb{R}^d$. Abstractly, each $v \in \mathcal{P}_w$ should be derivable from the online state $(B, A_w, R, B^{(w-1)})$ without access to F_w (so that F_w enters only through score evaluation, not through candidate construction), and the pool should cover the expected failure modes of the score so that the argmax can avoid them.

One concrete pool that has been validated empirically uses the five candidates below; the method is not tied to this specific choice.

- *Streaming-optimizer slot 2.* $v_{\text{opt}}^{(2)}$: the second direction returned by projected-gradient ascent on $\text{score}_w^{\text{add}}$ initialised orthogonally to the first-slot optimum.
- *Current-block SVD complement.* v_{blk} : the top right singular vector of A_w projected onto the orthogonal complement of the first-slot direction.
- *Sketched SVD complement.* v_{sk} : the top right singular vector of M_w projected onto the orthogonal complement of the first-slot direction.
- *Projected oracle analogue.* v_{q2} : a reference direction constructed by projecting a candidate onto the row space of the current block and renormalising (the specific construction depends on the available state; in the purely online setting, this candidate is replaced by a second SVD-complement variant).

- *Previous-window slot-2 carry.* v_{prev} : the slot-2 direction selected at window $w - 1$, carried forward as a stability anchor.

Pool edits that add candidates with disproportionately large $\|A_w v\|_2^2$ but negligible past or future support shift the argmax toward those candidates by inflating the score’s gain numerator, and should be validated with an ablation before deployment.

Stage 2 — Rayleigh–Ritz frame refinement. Let $Q_c \in \mathbb{R}^{d \times r}$ with $Q_c^\top Q_c = I$ and $\text{span}(Q_c) = \mathcal{S}_w$ (obtained from QR of the selected pool elements). Compute the restricted SVD

$$M_w Q_c = U_c \Sigma_c V_c^\top, \quad M_w = \begin{pmatrix} B \\ A_w \end{pmatrix}, \quad (9)$$

and take

$$\hat{V}_w := Q_c V_c. \quad (10)$$

$\hat{V}_w$ is the top- r right singular frame of M_w restricted to $\mathcal{S}_w$, equivalently the unique (up to sign) orthonormal frame of $\mathcal{S}_w$ that diagonalises $Q_c^\top M_w^\top M_w Q_c$. The output is independent of the choice of Q_c : a different ONB of $\mathcal{S}_w$ produces $Q_c R$ for some rotation R , which rotates V_c to $R^\top V_c$, and the product $Q_c V_c$ is unchanged.

By Rayleigh–Ritz, $\hat{V}_w$ is the minimiser of the Stage-2 frame objective:

$$\hat{V}_w = \arg \min_{V \in \text{St}(r, \mathcal{S}_w)} \|M_w - M_w V V^\top\|_F = \arg \max_{V \in \text{St}(r, \mathcal{S}_w)} \|M_w V\|_F^2, \quad (11)$$

where $\text{St}(r, \mathcal{S}_w)$ is the set of orthonormal $d \times r$ matrices with columns in $\mathcal{S}_w$.

Stage 3 — carry-sketch update. Using the Ritz frame $\hat{V}_w$, form

$$Y_w := M_w \hat{V}_w, \quad U_{,w} := \text{orth}(Y_w), \quad (12)$$

compute the SVD

$$U_{,w}^\top M_w = U_{\text{small},w} \Sigma_w V_w^\top, \quad (13)$$

and store

$$U_{\text{proj},w} = U_{,w} U_{\text{small},w}, \quad S_w = \Sigma_w, \quad V_w^\top \text{ (the right singular factor)}. \quad (14)$$

The sketch passed to the next window is

$$B^{(w)} := \Sigma_w V_w^\top. \quad (15)$$

Why this is the sketch carried into the next window. Consider the naive alternative of carrying $\widehat{V}_w$ together with column norms $s_i := \|M_w \widehat{v}_i\|_2$. With

$$M_w = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \widehat{V}_w = [e_1, e_2], \quad (16)$$

one has $s_1 = s_2 = 1$, so $B_{\text{diag}} := \text{diag}(s) \widehat{V}_w^\top = I$ gives $\|B_{\text{diag}} v\|_2^2 = 1$ uniformly on $\mathbb{S}^1$, whereas

$$\|M_w v\|_2^2 = v^\top \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} v \quad (17)$$

vanishes at $v = (1, -1)/\sqrt{2}$ and equals 2 at $v = (1, 1)/\sqrt{2}$. Column norms drop the cross-correlations $(M_w \widehat{v}_i)^\top (M_w \widehat{v}_j)$, which are generically nonzero because the Ritz-refined frame $\widehat{V}_w$ is aligned with the singular structure of M_w restricted to $\mathcal{S}_w$ rather than the full M_w . Carrying $B^{(w)} = \Sigma_w V_w^\top$ instead is the norm-preserving right-action form of $P_{U,w} M_w$, where $P_{U,w} = U_{,w} U_{,w}^\top$ is the orthogonal projector onto the retained left space. For any v , $\|B^{(w)} v\|_2 = \|P_{U,w} M_w v\|_2$, and for $v \in \mathcal{S}_w$ this equals $\|M_w v\|_2$ exactly since $M_w v \in \text{span}(U_{,w})$. Hence $B^{(w)\top} B^{(w)}$ reproduces the full Gram $\widehat{V}_w^\top M_w^\top M_w \widehat{V}_w$ on the retained right span, all off-diagonals included. The row sample $R^{(w)}$ enters the pooled entropy at the next window.

Because the Stage-2 Ritz frame already satisfies $(M_w \widehat{v}_i)^\top (M_w \widehat{v}_j) = \sigma_i^2 \delta_{ij}$, the Stage-3 carry-SVD receives an already-diagonalised input on $\mathcal{S}_w$; cross-correlations only appear through the interaction of $\mathcal{S}_w$ with its complement, which the carry-SVD handles exactly.

Combined entropy. With $y := C_w v$, the definition of h_2 gives

$$h_2(C_w v) = 4 \log \|y\|_2 - 4 \log \|y\|_4. \quad (18)$$

Applied to the stacked vector combining the old row sample Rv with the current window $A_w v$, rare large entries in either source contribute to the same correction factor $\phi_w(v)$; the reservoir and the current window are not separated in the entropy term.

Simplified form. Let

$$c_w := \frac{\log(\text{row}(N_w)/n)}{2 \log \text{row}(C_w)}, \quad y := C_w v. \quad (19)$$

Since

$$h_2(C_w v) = -\log \left(\frac{\|y\|_4^4}{\|y\|_2^4} \right), \quad (20)$$

the entropy multiplier is

$$\left(\frac{\text{row}(N_w)}{n}\right)^{-h_2(C_w v)/(2 \log \text{row}(C_w))} = \left(\frac{\|y\|_4^4}{\|y\|_2^4}\right)^{c_w} = \|C_w v\|_4^{4c_w} \|C_w v\|_2^{-4c_w}. \quad (21)$$

Thus

$$\text{score}_w^{\text{add}}(v) = (\|Bv\|_2^2 + \|A_w v\|_2^2) \phi_w(v), \quad \text{score}_w^{\text{HM}}(v) = n \cdot \text{HM}(\hat{\lambda}_{A_w}(v), \hat{\lambda}_{F_w}(v)) \cdot \phi_w(v), \quad (22)$$

with the common entropy factor

$$\phi_w(v) = \|C_w v\|_4^{4c_w} \|C_w v\|_2^{-4c_w}. \quad (23)$$

When $\text{row}(N_w) < n$, $c_w < 0$, and high-entropy pooled vectors are boosted.

Expanded row-sample form. Using

$$C_w^\top C_w v = R^\top R v + A_w^\top A_w v, \quad C_w^\top (C_w v)^{\odot 3} = R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}, \quad (24)$$

the entropy factor can be evaluated without explicitly forming C_w :

$$\phi_w(v) = \left(\frac{\|Rv\|_4^4 + \|A_w v\|_4^4}{(\|Rv\|_2^2 + \|A_w v\|_2^2)^2} \right)^{c_w}. \quad (25)$$

Gradient-style quantities on the entropy factor reuse these pooled expressions:

$$C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) = \frac{R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}}{\|Rv\|_4^4 + \|A_w v\|_4^4}, \quad \frac{C_w^\top C_w v}{\|C_w v\|_2^2} = \frac{R^\top R v + A_w^\top A_w v}{\|Rv\|_2^2 + \|A_w v\|_2^2}. \quad (26)$$

Stored-basis form. If

$$B = S V^\top, \quad V = [v_1, \dots, v_{r_{\text{sk}}}], \quad S = \text{diag}(s_1, \dots, s_{r_{\text{sk}}}), \quad (27)$$

then

$$\|Bv\|_2^2 = \sum_{j=1}^{r_{\text{sk}}} s_j^2 (v_j^\top v)^2, \quad B^\top B v = \sum_{j=1}^{r_{\text{sk}}} s_j^2 (v_j^\top v) v_j. \quad (28)$$

The past-energy term of the additive score and its gradient are therefore evaluated from (V, s) directly, without materialising B .

Gradient of the additive score. Let $y_B := Bv$, $z := A_w v$, $y := C_w v$. Then

$$\nabla (\|Bv\|_2^2 + \|A_w v\|_2^2) = 2B^\top Bv + 2A_w^\top A_w v, \quad (29)$$

and

$$\nabla \log \phi_w(v) = 4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2}. \quad (30)$$

The full Euclidean gradient is

$$\nabla \text{score}_w^{\text{add}}(v) = \text{score}_w^{\text{add}}(v) \left[\frac{2B^\top Bv + 2A_w^\top A_w v}{\|Bv\|_2^2 + \|A_w v\|_2^2} + 4c_w C_w^\top \left(\frac{(C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) - 4c_w \frac{C_w^\top C_w v}{\|C_w v\|_2^2} \right], \quad (31)$$

provided $\|Bv\|_2^2 + \|A_w v\|_2^2 > 0$.

Gradient of the HM score. Let $a(v) := \|A_w v\|_2^2$, $f(v) := \|F_w v\|_2^2$, $p := \text{row}(A_w)$, $q := \text{row}(F_w)$, and $D(v) := qa(v) + pf(v) + \varepsilon$. Then

$$\text{score}_w^{\text{HM}}(v) = \frac{2n a(v)f(v)}{D(v)} \cdot \phi_w(v), \quad (32)$$

and, with $\nabla a(v) = 2A_w^\top A_w v$, $\nabla f(v) = 2F_w^\top F_w v$,

$$\nabla \left(\frac{2n a(v)f(v)}{D(v)} \right) = \frac{2n}{D(v)} (f(v)\nabla a(v) + a(v)\nabla f(v)) - \frac{2n a(v)f(v)}{D(v)^2} (q\nabla a(v) + p\nabla f(v)). \quad (33)$$

The entropy factor has the same gradient as above. The full Euclidean gradient of $\text{score}_w^{\text{HM}}$ follows by the product rule; it is not used in the pool-based Stage 1 defined above, which evaluates the HM score on a finite set rather than optimising it continuously, but it is needed if the pool is replaced by a continuous optimiser.

Optimization problem for candidate construction. Certain candidates in $\mathcal{P}_w$ are themselves defined as solutions to a continuous optimisation; for those the feasible set is

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v = 0} \text{score}_w^{\text{add}}(v), \quad (34)$$

where the columns of Q are previously accepted directions.

Projected full-gradient ascent. Use the exact gradient of the additive score. Project it onto the feasible tangent space:

$$g_{\text{tan}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{add}}(v). \quad (35)$$

Then take

$$\tilde{v}^+ = v + \eta g_{\text{tan}}(v), \quad v^+ = \frac{(I - QQ^\top) \tilde{v}^+}{\|(I - QQ^\top) \tilde{v}^+\|_2}, \quad (36)$$

with $\eta > 0$ chosen by backtracking line search on the true score.

Basis-expansion interpretation. If one is building a local search subspace rather than taking a pure steepest-ascent step, the new direction added to the basis is the projected full gradient

$$g_{\text{basis}}(v) := (I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{add}}(v). \quad (37)$$

In a reduced-subspace version with local right basis $Z \in \mathbb{R}^{d \times q}$ ($Z^\top Z = I$), the expansion residual is

$$r_Z(v) := (I - ZZ^\top)(I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{add}}(v). \quad (38)$$

Expand the basis with the normalised residual $r_Z(v)/\|r_Z(v)\|_2$ when $\|r_Z(v)\|_2/\|(I - QQ^\top - vv^\top) \nabla \text{score}_w^{\text{add}}(v)\|_2$ exceeds a prescribed tolerance; optionally follow with Krylov refinements.

Rayleigh–Ritz optimality. The Stage-2 Ritz frame $\hat{V}_w$ provides the following standard guarantees inside $\mathcal{S}_w$ (no assumption beyond existence of the restricted SVD):

1. *Energy ordering.* $\|M_w \hat{V}_w[:, i]\|_2 = \sigma_i(M_w Q_c)$, ordered decreasing in i .
2. *Diagonalisation.* $\hat{V}_w^\top M_w^\top M_w \hat{V}_w = \text{diag}(\sigma_1^2, \dots, \sigma_r^2)$.
3. *Optimal truncation.* For every $k \leq r$, the first k columns of $\hat{V}_w$ minimise $\|M_w - M_w V V^\top\|_F$ over $V \in \text{St}(k, \mathcal{S}_w)$.

None of these hold generically for a Gram–Schmidt orthonormalisation of the selected pool elements: Gram–Schmidt preserves the span but leaves the columns aligned with the pool’s construction rather than with the singular structure of $M_w|_{\mathcal{S}_w}$.

Role of the score: test-space certificate. The additive score is a certified lower bound on $\|Av\|_2^2$ under (A1) + (A2). Applied to Stage 1 (S1), the selected candidates satisfy

$$\sum_{v \in \tilde{V}_w} \|Av\|_2^2 \gtrsim \sum_{v \in \tilde{V}_w} \text{score}_w^{\text{add}}(v) \geq \sum_{i=1}^r \text{score}_w^{\text{add}}(v_i^*) \gtrsim \sum_{i=1}^r \sigma_i(A)^2, \quad (39)$$

provided the top- r right singular vectors v_i^* of A are contained in (or well approximated by) the pool $\mathcal{P}_w$. The score’s role is therefore to supply a *test-space certificate*: the span $\mathcal{S}_w$ covers the top- r singular mass of A up to the (A1,A2) gap. Rayleigh–Ritz then extracts the provably-best rank- r frame inside $\mathcal{S}_w$, and the carry-sketch update stores the restricted operator’s Gram factorisation for the next window.

The HM variant substitutes a stationarity-based estimator for the certified bound at Stage 1. Under (A1’)+(A2) at $v = v_i^*$, the heuristic analogue $\text{score}_w^{\text{HM}}(v_i^*) \approx \sigma_i(A)^2 \phi_w(v_i^*)$ would give the same certificate, but the violation tolerance is not quantified.

[INCOMPLETE] Coverage certificate for the HM variant. For the Stage 1 certificate to carry over to the HM score, we need an analogue of $\text{score}_w^{\text{add}}(v_i^*) \gtrsim \sigma_i(A)^2$ for $\text{score}_w^{\text{HM}}$. A natural candidate is

$$\text{score}_w^{\text{HM}}(v_i^*) \gtrsim \sigma_i(A)^2 \quad \text{under (A1')+(A2),} \quad (40)$$

which holds exactly when $\hat{\lambda}_{A_w}(v_i^*) \approx \hat{\lambda}_{F_w}(v_i^*) \approx \lambda(v_i^*) = \sigma_i(A)^2/n$, but the (A1')-violation tolerance of the inequality has not been quantified. What is needed:

- A quantitative stability bound of the form $|\text{score}_w^{\text{HM}}(v) - \|Av\|_2^2 \phi_w(v)| \leq g(\hat{\lambda}_{A_w}(v), \hat{\lambda}_{F_w}(v))$, so that the HM score's lower-bound gap is controlled by the stationarity deficit.
- A translation of such a stability bound into a subspace-distance bound $\text{dist}(\mathcal{S}_w, \text{span}(v_1^*, \dots, v_r^*))$, matching the role of the (A1) + (A2) gap for the additive score.

[INCOMPLETE] End-to-end approximation bound. The three-stage algorithm combines a bound-backed Stage 1 (additive score) with a theorem-backed Stage 2 (Rayleigh–Ritz) and a norm-preserving Stage 3 (carry sketch), but a single end-to-end bound on $\|A - AV_{\text{final}}V_{\text{final}}^\top\|_F$ in terms of windowed quantities is not written. The ingredients are in place:

- Stage-1 test-space coverage: $\|AP_{\mathcal{S}_w}\|_F^2 \gtrsim \sum_{v \in \tilde{V}_w} \text{score}_w(v) \gtrsim \sum_{i=1}^r \sigma_i(A)^2$ under (A1) + (A2).
- Stage-2 Ritz optimality inside $\mathcal{S}_w$.
- Stage-3 carry-sketch norm-preservation on the retained right span.

What is missing is a composition argument that tracks the per-window approximation error through the sequence of carry updates and yields a final bound in terms of the per-window (A1,A2) gaps and the Ritz truncation residuals $\|M_w Q_c - U_c \Sigma_c V_c^\top\|$ (zero for $r = \dim \mathcal{S}_w$, nonzero otherwise). For the HM variant, such a bound must further absorb the gap documented in the [INCOMPLETE] HM-derivation paragraph.

First-window special case. When $B = 0$, $R = \emptyset$, and F_1 is the first lookahead buffer, the entropy factor reduces to

$$\phi_1(v) = \|A_1 v\|_4^{4c_1} \|A_1 v\|_2^{-4c_1}, \quad c_1 = \frac{\log(\text{row}(A_1)/n)}{2 \log \text{row}(A_1)}. \quad (41)$$

The additive score reduces to

$$\text{score}_1^{\text{add}}(v) = \|A_1 v\|_2^2 \phi_1(v) = \|A_1 v\|_2^2 \|A_1 v\|_4^{4c_1} \|A_1 v\|_2^{-4c_1}, \quad (42)$$

and the HM score to

$$\text{score}_1^{\text{HM}}(v) = \frac{2n \|A_1 v\|_2^2 \|F_1 v\|_2^2}{\text{row}(F_1) \|A_1 v\|_2^2 + \text{row}(A_1) \|F_1 v\|_2^2 + \varepsilon} \cdot \phi_1(v). \quad (43)$$

The three-stage structure is the same as for general w ; only $M_1 = A_1$ differs in that it has no sketch block.

Practical remarks.

- The objective is nonconvex in candidates that are themselves continuously optimised; multiple restarts are useful for those candidates. Reject or regularise steps when $\|C_w v\|_2$, $\|C_w v\|_4$, $\|Bv\|_2^2 + \|A_w v\|_2^2$, $\|A_w v\|_2$, or $\|F_w v\|_2$ is numerically tiny in a formula that divides by that quantity.
- The expanded row-sample form avoids forming C_w explicitly; the pooled norms must still include both Rv and $A_w v$.
- The row sample is not a replacement for the old sketch. B supplies retained old energy; R supplies raw old rows for the pooled entropy. A pure reservoir gives coverage but can miss rare large rows that dominate ℓ_4 terms. A practical row sample should combine a uniform reservoir with a small set of rows having large entries on current sketch directions.
- The Rayleigh–Ritz reframe is numerically well-conditioned even when selected pool elements are nearly collinear in $\mathbb{R}^d$: the SVD sees the rank-deficiency through $\sigma_r(M_w Q_c) \rightarrow 0$, whereas a Gram–Schmidt orthonormalisation of the same elements would amplify the residual direction through division by a small norm. Fall back to a smaller effective rank only if $\sigma_r(M_w Q_c)$ is below a prescribed tolerance.
- The online HM variant requires a lookahead lag of $\text{row}(F_w)$ rows. It is online-deployable at a latency cost but not strictly streaming. The additive score with $F_w = \emptyset$ remains strictly streaming (zero lag).
- The reservoir R enters both scores only through the entropy factor ϕ_w , not through the gain factors.

Open items. The [INCOMPLETE] paragraphs above flag the missing pieces of the theoretical story:

1. Derivation of $\text{score}_w^{\text{HM}}$ as a lower-bound quantity starting from a subsampling inequality analogous to $(\star)$.
2. A coverage certificate of the form $\text{score}_w^{\text{HM}}(v_i^*) \gtrsim \sigma_i(A)^2$ with quantified $(A1')$ -deficit tolerance.
3. End-to-end approximation bound composing Stage-1 coverage, Stage-2 Ritz optimality, and Stage-3 carry-sketch exactness across $w = 1, \dots, W$.

Filling (1) and (2) is required before the HM variant can be reported with the same theoretical status as the additive score. Filling (3) would promote the three-stage pipeline from a justified heuristic into an approximation algorithm with explicit error guarantees.